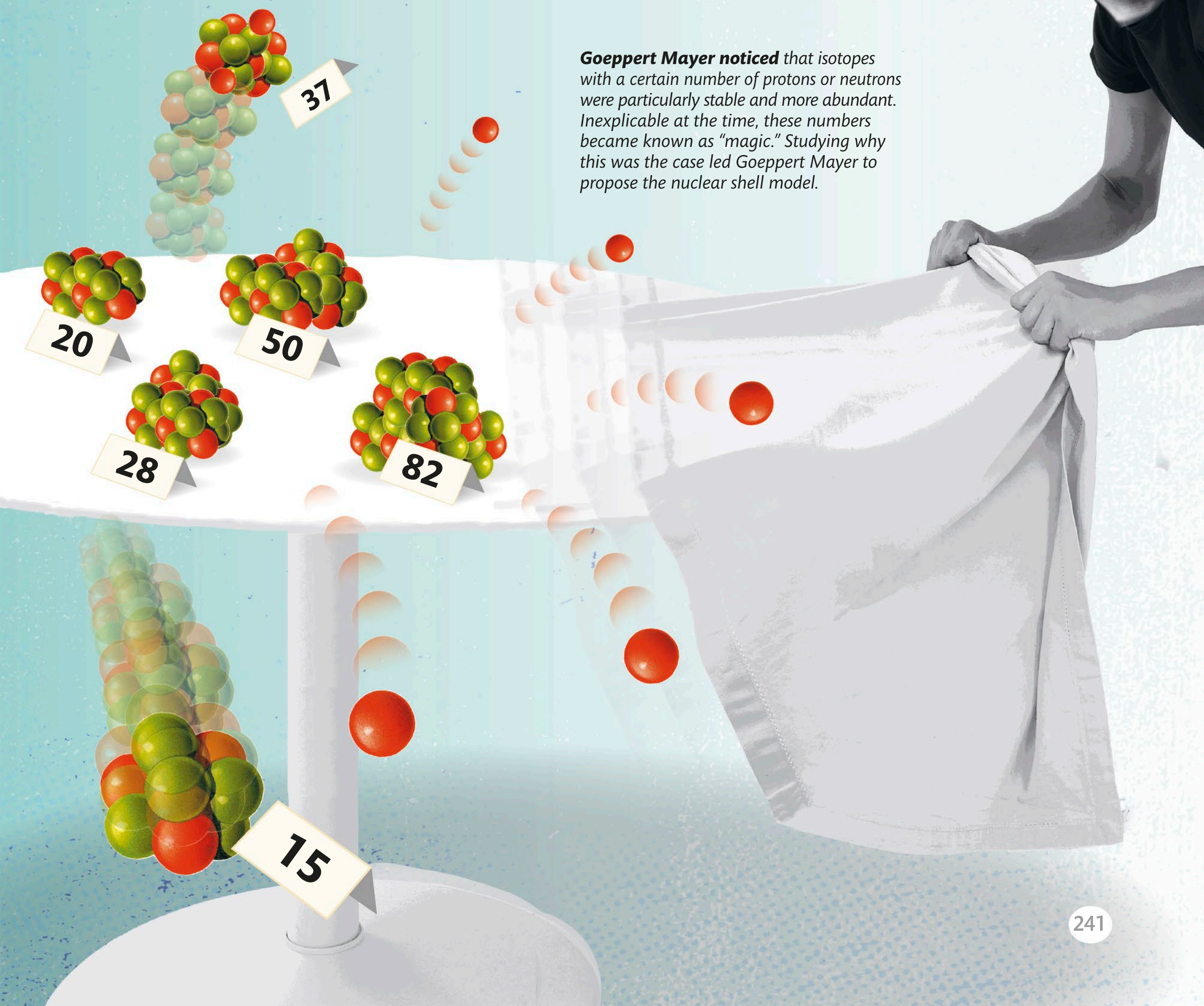
understanding of quantum physics had enabled her to recognize that spin-orbit coupling was the key to the correct solution: a breakthrough that was hugely important for nuclear physics.

In 1963, Goeppert Mayer was awarded the Nobel Prize in Physics in 1963 with Hans Jensen, who had come up with the same nuclear shell model independently, and Eugene Wigner (see box).

> **“**Winning the [Nobel] Prize wasn’t half as exciting **as doing the work itself.”**
>
> **Maria Goeppert Mayer**



Goeppert Mayer noticed *that isotopes with a certain number of protons or neutrons were particularly stable and more abundant. Inexplicable at the time, these numbers became known as “magic.” Studying why this was the case led Goeppert Mayer to propose the nuclear shell model.*